01

PART 1

Why grids need storage

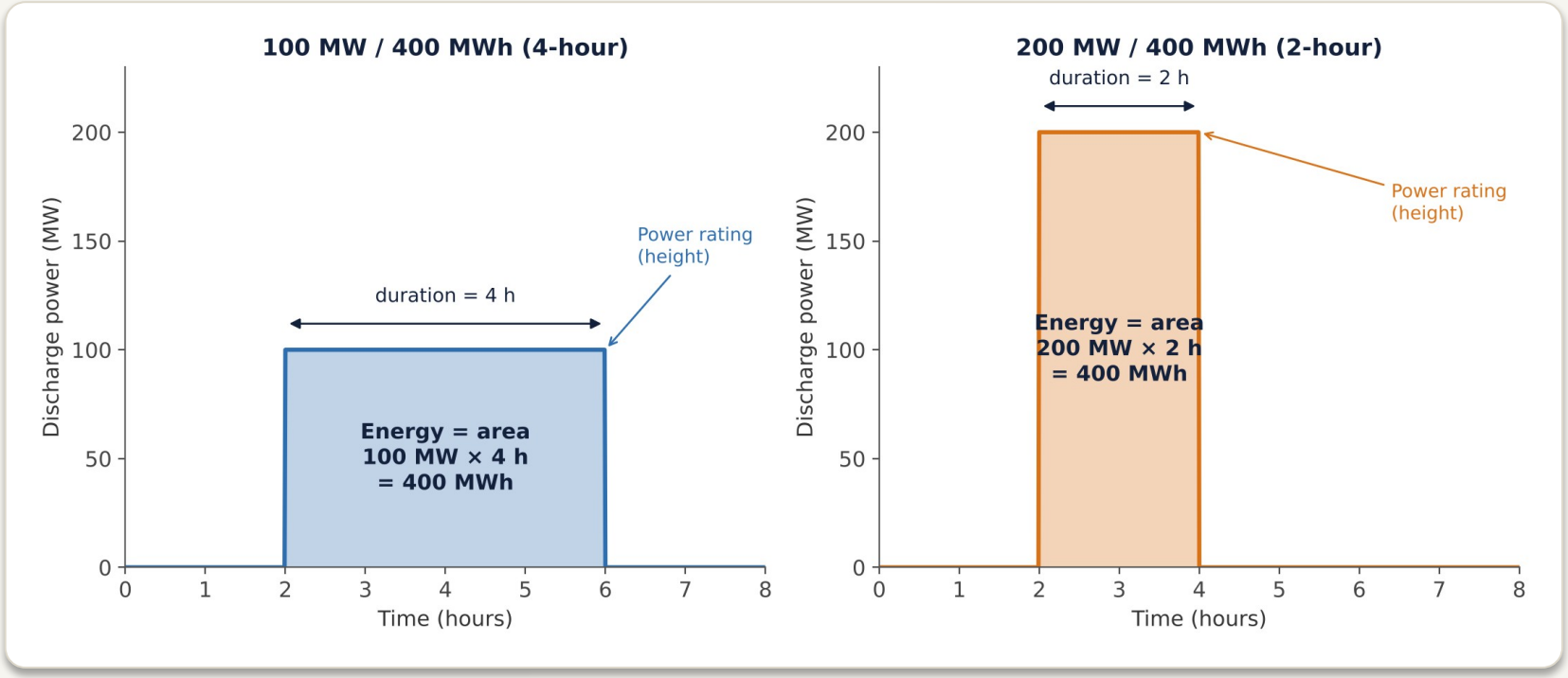
Electricity must be produced at the instant it is used. A battery breaks that rule: it moves energy in time and responds in milliseconds.

IN THIS PART

- Power (MW) versus energy (MWh), and duration
- The services a BESS sells, from milliseconds to hours
- Where storage sits: generation side, grid, customer side
- How BESS compares with other storage technologies

PART 1 · BASIC VOCABULARY

Power is the height, energy is the area



How to read this chart

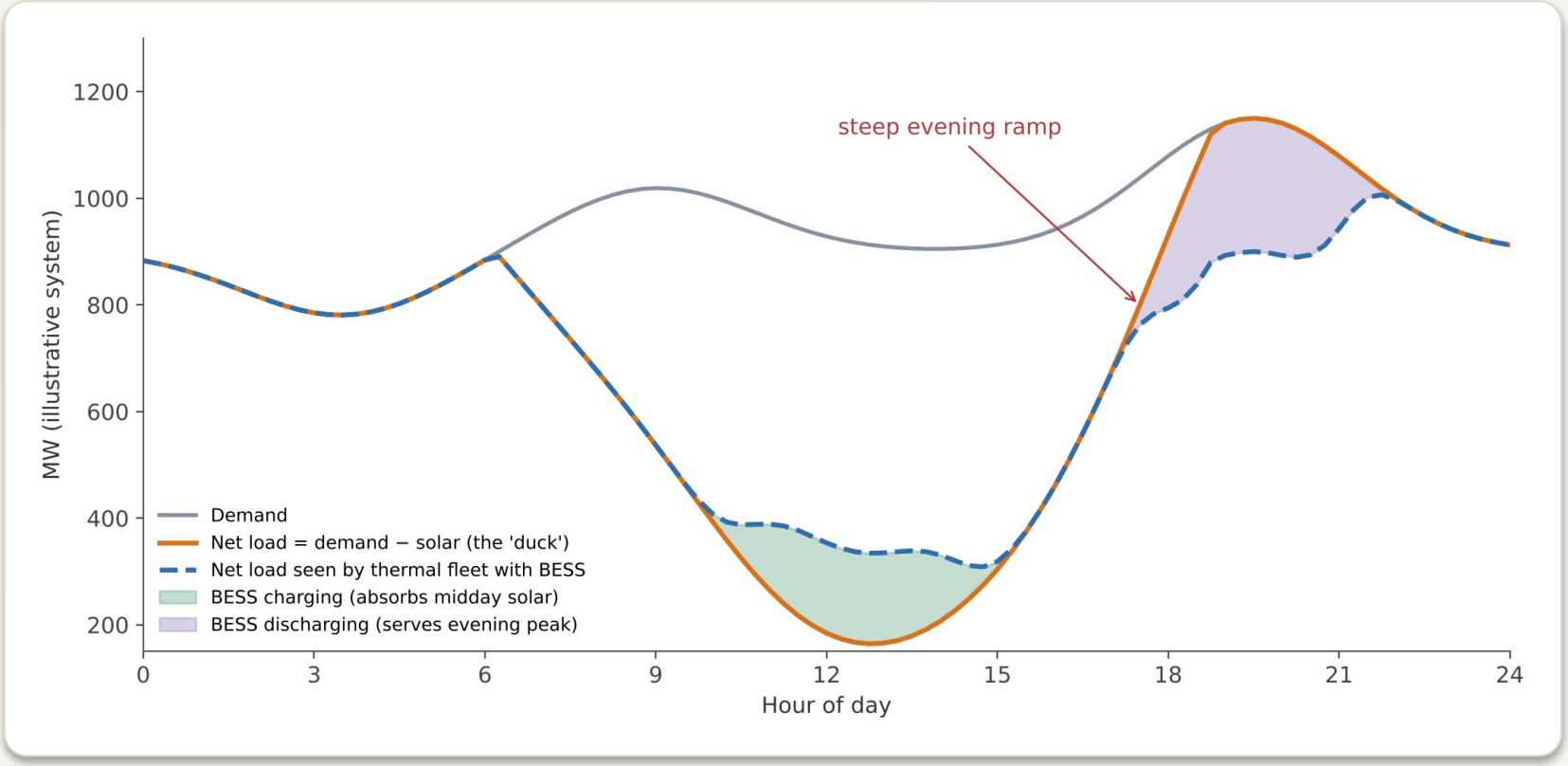
- **Power rating (MW):** how fast the plant can charge or discharge, set mainly by the inverters.
- **Energy capacity (MWh):** how much it can store, set mainly by the number of cells.
- **Duration (h) = energy ÷ power.** Both plants store 400 MWh, but one delivers it in 4 hours and the other in 2.

> Always quote a BESS as power / energy, e.g. 100 MW / 400 MWh (a ‘4-hour’ system).

MW = megawatt · MWh = megawatt-hour · BESS = battery energy storage system

PART 1 · THE GRID PROBLEM

Solar creates a midday surplus and a steep evening ramp



How to read this chart

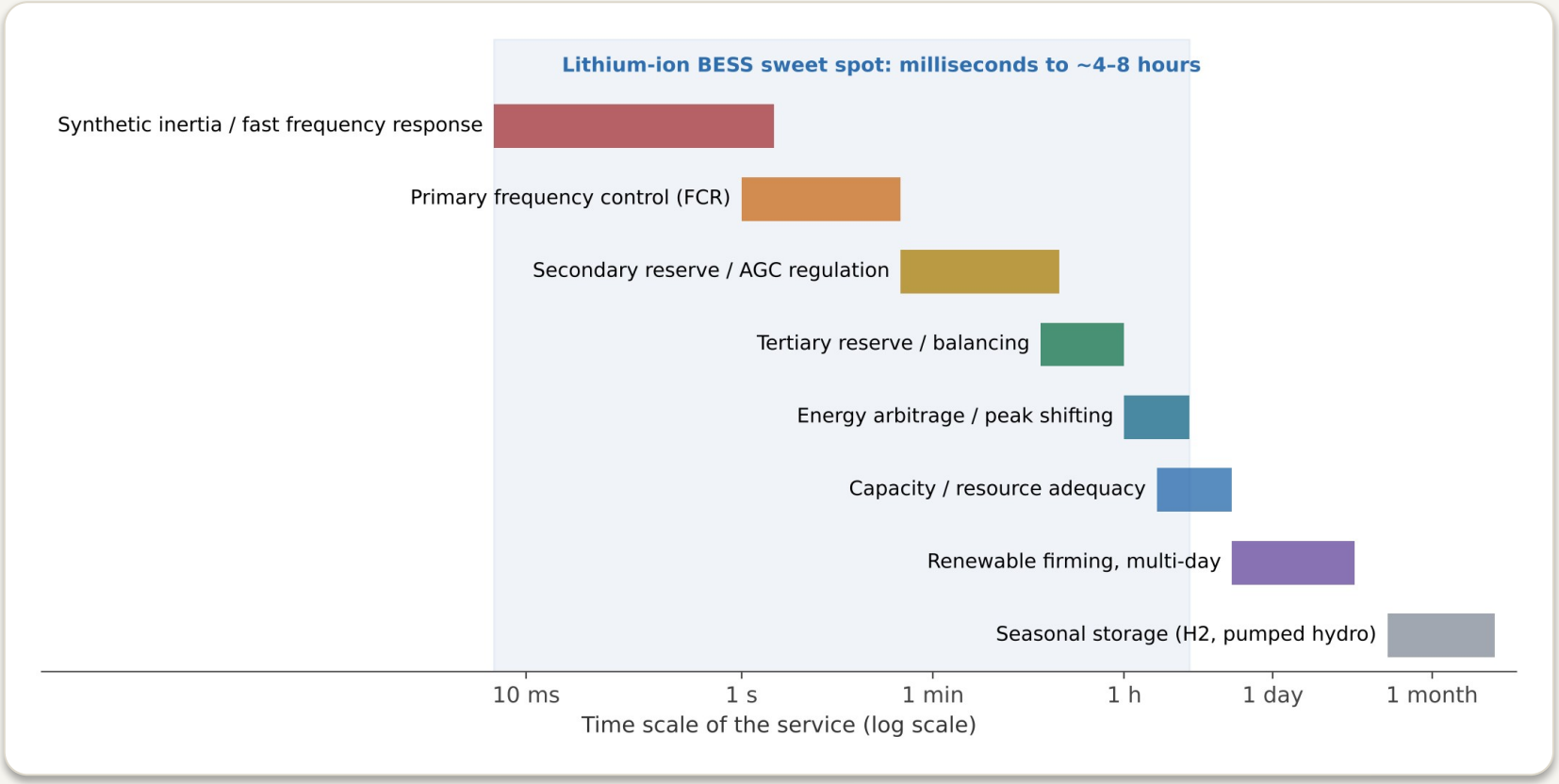
- **Grey:** customer demand through the day.
- **Orange:** net load = demand minus solar. Its shape is nicknamed the ‘duck curve’.
- **Green area:** the BESS charges from cheap midday solar.
- **Purple area:** the BESS discharges into the evening peak, flattening the ramp (blue dashed

A 4-hour battery moves midday solar into the evening: arbitrage, firming and peak capacity in one picture.

Illustrative system; synthetic data

PART 1 · SERVICES

One asset, services from milliseconds to hours



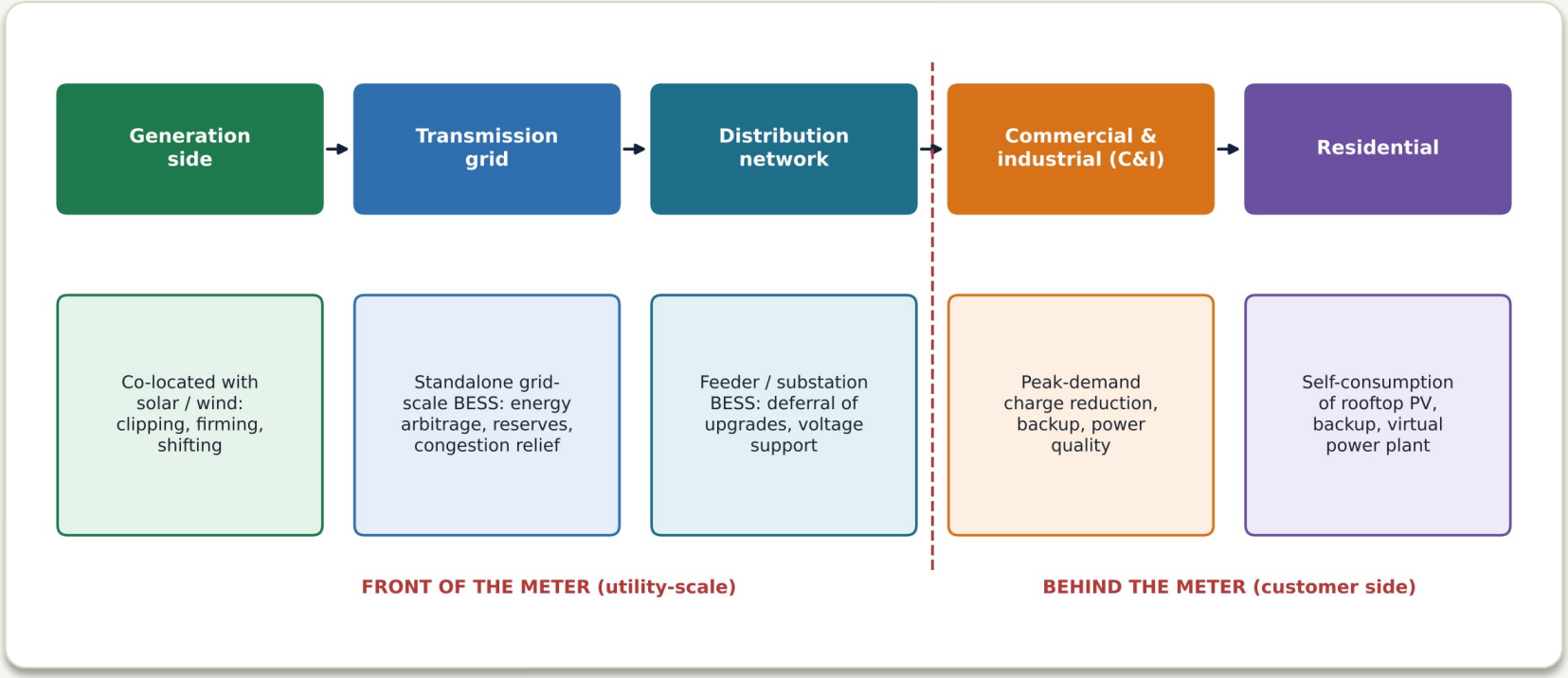
How to read this chart

- **Bars:** the time scale on which each grid service acts (log scale).
- **Shaded band:** where a lithium-ion BESS is technically and economically strong.
- **Left (fast):** inertia-like and frequency services, where response speed matters more than energy.
- **Right (slow):** multi-day and seasonal storage, where cheap energy capacity matters.

FCR = frequency containment reserve · AGC = automatic generation control · H2 = hydrogen

PART 1 · WHERE STORAGE SITS

Front of the meter and behind the meter



How to read this diagram

- **Top row:** the chain from generation to the end customer.
- **Bottom row:** the main job of a BESS at each location.
- **Red dashed line:** the customer's revenue meter. Left of it is 'front of the meter' (utility-scale); right is 'behind the meter'.

C&I = commercial and industrial · VPP = virtual power plant

PART 1 · SERVICES IN DETAIL

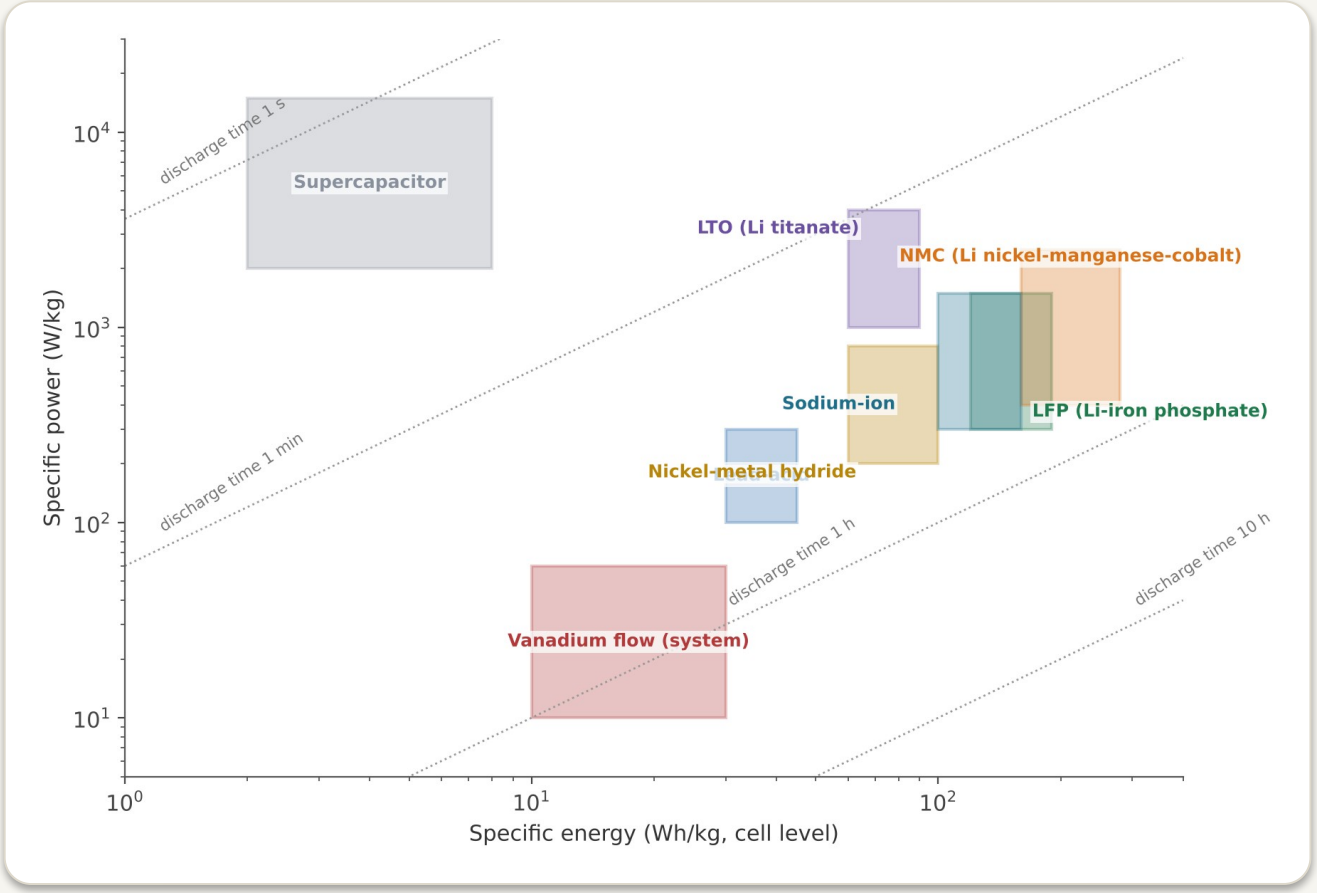
What each service asks of the battery

Service	What the battery does	Response	Typical duration	Key parameter
Fast frequency response / synthetic inertia	Injects power in proportion to frequency deviation or its rate of change	< 0.5 s	seconds	Power, response time
Primary frequency control (FCR, PFR)	Droop response to frequency deviation	< 30 s	15–30 min of energy	Power, SoC headroom
Secondary reserve / regulation (aFRR, AGC)	Follows a set-point sent every 2–4 s by the operator	30 s – 5 min	continuous	Ramp, cycling tolerance
Energy arbitrage / time shift	Charges at low price, discharges at high price	minutes	1–4 h	Energy, RTE, degradation
Capacity / adequacy	Available at the system peak	minutes	2–8 h (rules vary)	Sustained energy
Renewable firming, clipping capture	Smooths and shifts PV or wind output	seconds–hours	1–4 h	Coupling, energy
Voltage support, black start, congestion relief	Reactive power; energises a dead grid; relieves a line	ms–minutes	varies	Inverter rating, grid-forming

FCR/PFR = frequency containment / primary frequency response · aFRR = automatic frequency restoration reserve · RTE = round-trip efficiency

PART 1 · TECHNOLOGY COMPARISON

The Ragone plot: energy versus power density



How to read this chart

- **Horizontal axis:** specific energy, how much energy per kilogram.
- **Vertical axis:** specific power, how fast per kilogram.
- **Dotted diagonals:** constant discharge time. Moving right and down means longer discharge.
- **Lithium-ion** sits in the middle: high energy and high power at the same time.

Cell-level ranges, rounded from literature; indicative only

PART 1 · TECHNOLOGY COMPARISON

Storage technologies at a glance

Technology	Typical duration	Round-trip efficiency	Response	Life	Notes
Lithium-ion BESS (LFP)	0.5–8 h	85–90 % (AC)	< 100 ms	6,000–10,000 cycles, 15–20 yr	Dominant for new grid storage
Pumped hydro storage	6–24 h	70–80 %	seconds–minutes	50–100 yr	Needs geography; long build
Compressed air (CAES)	4–24 h	40–70 %	minutes	30–40 yr	Needs caverns
Vanadium redox flow	4–12 h	65–75 %	< 1 s	> 15,000 cycles, 20+ yr	Power and energy sized separately
Sodium-sulphur (NaS)	6 h	75–80 %	ms	~4,500 cycles	Runs at about 300 °C
Hydrogen (electrolyser + turbine/fuel cell)	days–seasons	30–40 %	s–min	20+ yr	Low efficiency; huge energy capacity
Flywheel / supercapacitor	seconds–minutes	85–95 %	ms	> 100,000 cycles	Power-only services

LFP = lithium iron phosphate · CAES = compressed air energy storage · NaS = sodium–sulphur